

Advanced Linux Users and Group Management with Policies

Types Of Users

Linux operating systems support different types of users, each serving a unique role and having different levels of permissions. Let's break down the concepts of **regular users**, **system users**, and the **root user**, along with command examples to create and manage them, from basic to advanced scenarios.

1. Regular User

A regular user is a standard user account that is created for individuals who log in to the system to perform daily activities. Regular users have restricted permissions and cannot perform administrative tasks unless explicitly granted with tools like **sudo**.

Characteristics:

- Home directory is typically located at `/home/username`.
- Limited access to system files and settings.
- Cannot run administrative commands unless given **sudo** privileges.

Basic Commands:

Create a Regular User:

```
sudo useradd -m username
```

Set a Password for the User:

```
sudo passwd username
```

Check the User's Details:

```
getent passwd username
```

This command (Get Entries) will display the user's information, including the assigned shell, home directory, and user ID (UID).

Intermediate Commands:

Create a User with a Specific Shell:

```
sudo useradd -m -s /bin/bash username
```

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- `-s /bin/bash`: Sets the shell to `/bin/bash` (can be changed to other shells like `/bin/sh` or `/bin/zsh`).

Assign the User to a Group:

Regular users are often part of a primary group. You can also add them to supplementary groups.

```
sudo usermod -aG groupname username
```

Advanced Commands:**Create a User with Expiration Date:**

```
sudo useradd -m -e YYYY-MM-DD username
```

- `-e`: Specifies the account's expiration date.

Create a User with Custom User ID (UID):

```
sudo useradd -m -u 2000 username
```

- `-u 2000`: Sets the UID to 2000 (useful for UID management in multi-system environments).

Delete a Regular User:

```
sudo userdel -r username
```

- `-r`: Deletes the user's home directory and mail spool.

2. System User

System users are typically created to run background services or daemons such as web servers, databases, and system services. These accounts are not meant for human interaction and often have limited or no shell access.

Characteristics:

- UIDs for system users are usually below 1000 (on most distributions).
- Often do not have home directories unless explicitly created.
- Shell is typically set to `/usr/sbin/nologin` or `/bin/false` to prevent interactive login.

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Basic Commands:

Create a Basic System User:

```
sudo useradd -r sysuser
```

- `-r`: Creates a system account with a UID below 1000.

Assign a Shell That Prevents Login:

```
sudo useradd -r -s /usr/sbin/nologin sysuser
```

- `-s /usr/sbin/nologin`: Assigns a shell that prevents login. If `sysuser` attempts to log in, the system will show `This account is currently not available.`

Intermediate Commands:

Create a System User with a Custom Home Directory:

```
sudo useradd -r -m -d /opt/sysuser sysuser
```

- `-d /opt/sysuser`: Sets the custom home directory at `/opt/sysuser`.

Set Password for a System User (Optional):

```
sudo passwd sysuser
```

This command is only relevant if the system user needs to authenticate for some tasks.

Advanced Commands:

Modify an Existing System User's Shell:

```
sudo usermod -s /bin/bash sysuser
```

- Changes the shell for `sysuser` to `/bin/bash` (useful if you need to temporarily enable interactive login).

Lock/Unlock a System User Account:

- **Lock the Account:**

```
sudo passwd -l sysuser
```

- Prevents the user from logging in even if they have a shell assigned.

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➤ **Unlock the Account:**

```
sudo passwd -u sysuser
```

Main Differences b/w System User and Regular User:

System User:

- Created primarily for managing and running system services, processes, and applications.
- Typically used by the operating system to handle background tasks like web servers, databases, and other daemons.

Regular User:

- Created for human users to perform general computing tasks such as file management, software execution, and system interaction.
- Regular users have home directories and can log in to the system.

3. Root User

The root user is the **superuser** in a Linux system with **unrestricted access**. The root user can perform all administrative tasks, such as installing software, managing system services, and changing file permissions.

Characteristics:

- Has a UID of 0.
- Has complete control over the system.
- Can modify any file, change user permissions, and execute any command.

Accessing the Root User:

Switch to the Root User:

```
sudo su -
```

- This command opens a root shell.

Direct Login (If enabled):

If direct root login is enabled (not recommended for security reasons):

```
su -
```

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To avoid logging in directly as `root`, it is recommended to create a user with `sudo` privileges:

Create a User and Add to `sudo` Group:

```
sudo useradd -m -s /bin/bash adminuser  
sudo usermod -aG sudo adminuser
```

- `-aG sudo`: Adds `adminuser` to the `sudo` group, giving it root privileges.

Verify Root Privileges:

Log in as the user and check with:

```
sudo -l
```

This will show a list of commands the user can run with `sudo`.

Advanced Root User Management:**Set Up `sudo` Without Password:**

Edit the `/etc/sudoers` file (use `visudo` for safe editing):

```
sudo visudo
```

Add the following line for a specific user:

```
adminuser ALL=(ALL) NOPASSWD:ALL
```

Restrict `sudo` Commands:

Modify the `/etc/sudoers` file to limit which commands a user can execute as root:

```
adminuser ALL=(ALL) /usr/bin/systemctl, /usr/bin/reboot
```

Deleting a User with Root Privileges:

If you need to delete a user who has been granted root privileges:

Remove the User:

```
sudo userdel -r adminuser
```

- Deletes the user and its home directory.

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Revoke sudo Access:

Remove the user from the `sudo` group:

```
sudo deluser adminuser sudo
```

Types of Groups

In Linux, user accounts can belong to one or more groups, which help manage permissions and access control for files and directories. Understanding primary and secondary (or supplementary) groups is essential for effective user management.

1. Primary Group

- **Definition:** The primary group is the default group assigned to a user account. It is the group that the user is associated with when they create new files or directories. By default, a user's primary group has the same name as the user.
- **User Creation:** When a user is created, a primary group is automatically assigned to them.

Basic Commands for Managing Primary Groups

Create a New User with a Primary Group:

When you create a new user with the `useradd` command, you can specify the primary group using the `-g` option:

```
sudo useradd -m -g groupname username
```

- `-g groupname`: Sets the specified group as the primary group for the user.

Check User's Primary Group:

To view a user's primary group, you can use the `id` command:

```
id username
```

Output example:

```
uid=1001(username) gid=1001(groupname) groups=1001(groupname)
```

Change a User's Primary Group:

You can change a user's primary group using the `usermod` command:

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```
sudo usermod -g newgroup username
```

2. Secondary (Supplementary) Groups

- **Definition:** Secondary groups (also known as supplementary groups) are additional groups that a user can belong to, beyond their primary group. These groups grant the user additional permissions to access files and resources associated with those groups.
- **File Permissions:** When a user creates a file, the primary group is the group that will own the file. However, secondary groups allow users to access files and directories owned by those groups.

Basic Commands for Managing Secondary Groups

Create a New User with Secondary Groups:

You can specify supplementary groups when creating a user with the `-G` option:

```
sudo useradd -m -g primarygroup -G secondarygroup1, secondarygroup2 username
```

Add a User to Secondary Groups:

To add an existing user to one or more secondary groups:

```
sudo usermod -aG groupname username
```

Check a User's Groups:

To see all groups that a user belongs to (including primary and secondary groups):

```
groups username
```

Or:

```
id username
```

Remove a User from a Secondary Group:

To remove a user from a secondary group, you can use the `gpasswd` command:

```
sudo gpasswd -d username groupname
```

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Advanced Concepts

Understanding Group ID (GID)

- Each group is assigned a unique Group ID (GID). The primary group of a user has a GID associated with it.
- You can view group information using:

```
getent group groupname
```

Group Permissions and Access Control

- File permissions are influenced by the user's primary and supplementary groups.
- **Permissions:** Each file has three types of permissions: read (r), write (w), and execute (x) for the owner, group, and others.

Setting Permissions

To set permissions for a group on a file or directory, you can use the `chmod` command:

```
chmod g+rxw filename    # Grants read, write, and execute
permissions to the group.
```

```
chmod g-w filename      # Removes write permission from the group.
```

Changing Group Ownership

You can change the group ownership of a file or directory using the `chown` command:

```
sudo chown :groupname filename
```

Access Control Lists (ACLs) in Linux

Introduction to ACLs:

- ACLs allow you to define more granular permissions for files and directories beyond the standard owner/group/others model.
- They can specify permissions for individual users or groups.

Basic Terminology:

- **User (u):** An individual user.
- **Group (g):** A group of users.
- **Other (o):** All other users not covered by the above.

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- **Mask (m):** Defines the maximum permissions for users and groups.

Common Options

- **-m:** Modify the ACL.
- **-x:** Remove an ACL entry.
- **-b:** Remove all ACL entries.
- **-k:** Remove the default ACL.
- **-R:** Apply changes recursively to directories and their contents.
- **-n:** Don't resolve user/group names.

Check if ACL is enabled on the filesystem:

```
mount | grep acl
```

- If ACL is not enabled, use the following command to remount (assuming the filesystem is ext4):

```
sudo mount -o remount,acl /
```

Create Sample Files and Directories:

```
mkdir example_dir  
touch example_dir/file1.txt  
touch example_dir/file2.txt
```

Set Default Permissions:

- Set the default permissions for the directory:

```
chmod 770 example_dir
```

View Current ACLs:

- Use `getfacl` to view current permissions:

```
getfacl example_dir
```

Modify ACLs:

- Add a user with specific permissions:

```
setfacl -m u:username:rw example_dir/file1.txt
```

- Example for adding user `alice`:

```
setfacl -m u:alice:rw example_dir/file1.txt
```

Add Group Permissions:

- Add a group with read and execute permissions:

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```
setfacl -m g:groupname:rx example_dir/file2.txt
```

- Example for adding group `devs`:

```
setfacl -m g:devs:rx example_dir/file2.txt
```

Set Default ACLs for a Directory:

- Set default ACLs so that new files inherit these permissions:

```
setfacl -d -m u:alice:rwX example_dir
```

Remove an ACL Entry:

- To remove a specific ACL entry:

```
setfacl -x u:alice example_dir/file1.txt
```

Verify ACLs After Modifications:

- Check the updated ACLs:

```
getfacl example_dir
```

Practice Questions**Basic Questions****1. Regular User Creation**

- What command would you use to create a new regular user named `ali`?
- After creating a user, how do you set a password for that user?

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo useradd ahs
```

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo passwd ahs
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
```

2. User Information

- How can you check the details of a user named `baqar`?

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ id ahs
uid=1001(ahs) gid=1001(ahs) groups=1001(ahs),1004(project_a),1005(project_b)
```

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- Write the command to create a user `shakir` with `/bin/bash` as their default shell.
- If you want to add `shakir` to the group `developers`, what command would you use?

```
root@abdullah-sohail-VMware-Virtual-Platform:/home/abdullah-sohail# useradd -s /bin/bash sohail
root@abdullah-sohail-VMware-Virtual-Platform:/home/abdullah-sohail# usermod -aG developers sohail
usermod: group 'developers' does not exist
```

4. User Deletion

- What command would you use to delete a user named `daniyal` along with their home directory?

```
root@abdullah-sohail-VMware-Virtual-Platform:/home/abdullah-sohail# id sohail
uid=1004(sohail) gid=1006(sohail) groups=1006(sohail)
root@abdullah-sohail-VMware-Virtual-Platform:/home/abdullah-sohail# userdel -r sohail
userdel: sohail mail spool (/var/mail/sohail) not found
userdel: sohail home directory (/home/sohail) not found
root@abdullah-sohail-VMware-Virtual-Platform:/home/abdullah-sohail# id sohail
id: 'sohail': no such user
```

5. Understanding User Types

- Describe the main difference between a regular user and a system user in Linux.

Difference Between Regular User and System User

Feature	Regular User	System User
Purpose	Used by humans to log in and perform tasks	Used by the system or services (like web server, database)
Login	Can usually log in via terminal or GUI	Typically cannot log in (no shell or restricted shell)
UID Range	Usually 1000 and above	Usually below 1000 (like 0–999)
Home Directory	Usually has a home directory like <code>/home/username</code>	May have no home directory or a system directory like <code>/var/lib/service</code>
Examples	<code>alice</code> , <code>bob</code> , <code>ahs</code>	<code>www-data</code> , <code>sshd</code> , <code>mysql</code>

The University of Lahore**Intermediate Questions****6. Creating System Users**

- How would you create a system user named `webservice` that does not have login access?
- What command would you use to assign a shell that prevents login to an existing system user `sysuser`?

```
root@abdullah-sohail-VMware-Virtual-Platform:/home/abdullah-sohail# usermod -s /usr/sbin/nologin alice
root@abdullah-sohail-VMware-Virtual-Platform:/home/abdullah-sohail# grep alice /etc/passwd
alice:x:1002:1002::/home/alice:/usr/sbin/nologin
root@abdullah-sohail-VMware-Virtual-Platform:/home/abdullah-sohail#
```

7. Managing Root User Access

- What is the command to switch to the root user in a terminal?
- Describe how to create a new user `adminuser` and grant them `sudo` privileges.

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo su
[sudo] password for abdullah-sohail:
root@abdullah-sohail-VMware-Virtual-Platform:/home/abdullah-sohail#
```

```
root@abdullah-sohail-VMware-Virtual-Platform:/home/abdullah-sohail# useradd -m ahs
```

```
root@abdullah-sohail-VMware-Virtual-Platform:/home/abdullah-sohail# usermod -aG sudo ahs
```

8. Primary and Secondary Groups

- When creating a new user `eve`, how do you set `admins` as her primary group and `developers` as a secondary group?
- What command would you use to check the groups to which user `fakhar` belongs?

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo groupadd admins
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo groupadd developers
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo useradd -m -g admins -G developers abdullah
```

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo passwd abdullah
New password:
BAD PASSWORD: The password fails the dictionary check - it is too simplistic/systematic
Retype new password:
passwd: password updated successfully
```

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```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ groups ahs
ahs : ahs project_a project_b
```

Advanced Questions**9. Advanced User Management**

- Write a command to create a new user `guest` that will expire on December 31, 2024.
- How can you change the user ID (UID) of an existing user `wasif` to 1500?

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo useradd -e 2026-03-03 abdullah
```

```
root@abdullah-sohail-VMware-Virtual-Platform:/home/abdullah-sohail# usermod -u 1500 alice
```

10. ACL Management

- What command would you use to set the ACL on a directory `project` to allow user `john` read and write access?
- How do you view the current ACL of a file named `data.txt`?

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo -i
root@abdullah-sohail-VMware-Virtual-Platform:~# mkdir /root/acl_test
root@abdullah-sohail-VMware-Virtual-Platform:~# cd /root/acl_test
root@abdullah-sohail-VMware-Virtual-Platform:~/acl_test# touch data.txt
root@abdullah-sohail-VMware-Virtual-Platform:~/acl_test# setfacl -m u:abdullah-sohail:rw data.txt
root@abdullah-sohail-VMware-Virtual-Platform:~/acl_test# getfacl data.txt
# file: data.txt
# owner: root
# group: root
user::rw-
user:abdullah-sohail:rw-
group::r--
mask::rw-
other::r--

root@abdullah-sohail-VMware-Virtual-Platform:~/acl_test#
```

11. Group Management

- How can you remove a user `george` from the secondary group `marketing`?
- Write the command to change the group ownership of the file `report.txt` to `finance`.

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```
root@abdullah-sohail-VMware-Virtual-Platform:/home/abdullah-sohail# gpasswd -d alice
Usage: gpasswd [option] GROUP
```

Options:

-a, --add USER	add USER to GROUP
-d, --delete USER	remove USER from GROUP
-h, --help	display this help message and exit
-Q, --root CHROOT_DIR	directory to chroot into
-r, --remove-password	remove the GROUP's password
-R, --restrict	restrict access to GROUP to its members
-M, --members USER,...	set the list of members of GROUP
--extrausers	use the extra users database
-A, --administrators ADMIN,...	set the list of administrators for GROUP

Except for the -A and -M options, the options cannot be combined.

```
root@abdullah-sohail-VMware-Virtual-Platform:/home/abdullah-sohail# chown :alice report.txt
```

12. Advanced Sudo Configuration

- Explain how to edit the `/etc/sudoers` file to allow `adminuser` to run the `systemctl` command without being prompted for a password.
- How can you restrict `adminuser` to only restart the web server using `sudo`?

```
root@abdullah-sohail-VMware-Virtual-Platform:/home/abdullah-sohail# visudo
```

```
GNU nano 7.2 /etc/sudoers
#
# This file MUST be edited with the 'visudo' command as root.
#
# Please consider adding local content in /etc/sudoers.d/ instead of
# directly modifying this file.
#
# See the man page for details on how to write a sudoers file.
#
Defaults        env_reset
Defaults        mail_badpass
Defaults        secure_path="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/snap/bin"
adminuser ALL=(ALL) NOPASSWD: /bin/systemctl
# This fixes CVE-2005-4890 and possibly breaks some versions of kdesu
# (#1011624, https://bugs.kde.org/show_bug.cgi?id=452532)
Defaults        use_pty

# This preserves proxy settings from user environments of root
# equivalent users (group sudo)
#Defaults:%sudo env_keep += "http_proxy https_proxy ftp_proxy all_proxy no_proxy"

# This allows running arbitrary commands, but so does ALL, and it means
# different sudoers have their choice of editor respected.
#Defaults:%sudo env_keep += "EDITOR"

# Completely harmless preservation of a user preference.
#Defaults:%sudo env_keep += "GREP_COLOR"
```

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```
GNU nano 1.2 /etc/sudoers
# This file MUST be edited with the 'visudo' command as root.
#
# Please consider adding local content in /etc/sudoers.d/ instead of
# directly modifying this file.
#
# See the man page for details on how to write a sudoers file.
#
Defaults                env_reset
Defaults                mail_badpass
Defaults                secure_path="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/snap/bin"
adminuser ALL=(ALL) NOPASSWD: /bin/systemctl restart apache2
# This fixes CVE-2005-4890 and possibly breaks some versions of kdesu
# (#1011624, https://bugs.kde.org/show_bug.cgi?id=452532)
Defaults                use_pty

# This preserves proxy settings from user environments of root
# equivalent users (group sudo)
#Defaults:%sudo env_keep += "http_proxy https_proxy ftp_proxy all_proxy no_proxy"

# This allows running arbitrary commands, but so does ALL, and it means
# different sudoers have their choice of editor respected.
#Defaults:%sudo env_keep += "EDITOR"

# Completely harmless preservation of a user preference.
#Defaults:%sudo env_keep += "GREP_COLOR"
```

Application Questions

13. Practical Scenarios

- You need to create a new user for a web application. This user should not have shell access but should be able to run a web server. Write down the commands you would use.
- A user reports that they cannot access a file in a directory. Explain how you would troubleshoot and resolve this issue, considering user permissions and group memberships.

```
root@abdullah-sohail-VMware-Virtual-Platform:/home/abdullah-sohail# useradd -r -s /usr/sbin/nologin webapp
root@abdullah-sohail-VMware-Virtual-Platform:/home/abdullah-sohail# mkdir /var/www/webapp
```

```
root@abdullah-sohail-VMware-Virtual-Platform:/home/abdullah-sohail# ls -l webapp
ls: cannot access 'webapp': No such file or directory
```

Troubleshoot file access problem:

```
root@abdullah-sohail-VMware-Virtual-Platform:/home/abdullah-sohail# ls -l webapp
total 0
root@abdullah-sohail-VMware-Virtual-Platform:/home/abdullah-sohail# id abdullah-sohail
uid=1000(abdullah-sohail) gid=1000(abdullah-sohail) groups=1000(abdullah-sohail),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),100(users),114(lpadmin)
root@abdullah-sohail-VMware-Virtual-Platform:/home/abdullah-sohail# ls -ld webapp
drwxr-xr-x 2 root root 4096 Mar  5 12:41 webapp
root@abdullah-sohail-VMware-Virtual-Platform:/home/abdullah-sohail#
```


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- Write a command to grant read and execute permissions to a group on a directory `sharedfolder`, and then remove write permission from that group.
- Explain how file permissions are inherited when a user creates a new file in a directory owned by a group.

```
root@abdullah-sohail-VMware-Virtual-Platform:/home/abdullah-sohail# chmod g+rx webapp
root@abdullah-sohail-VMware-Virtual-Platform:/home/abdullah-sohail# chmod g-w webapp
root@abdullah-sohail-VMware-Virtual-Platform:/home/abdullah-sohail#
```

Additional Questions**15. User and Group Troubleshooting**

- If a user cannot log in to the system, what steps would you take to diagnose the issue?
- Describe the process for changing the primary group of an existing user `zara` to `newgroup`.

```
root@abdullah-sohail-VMware-Virtual-Platform:/home/abdullah-sohail# cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
systemd-timesync:x:996:996:systemd Time Synchronization:/:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd:/bin/false
messagebus:x:101:101::/nonexistent:/usr/sbin/nologin
syslog:x:102:102::/nonexistent:/usr/sbin/nologin
systemd-resolve:x:991:991:systemd Resolver:/:/usr/sbin/nologin
uidd:x:103:103::/run/uidd:/usr/sbin/nologin
usbmux:x:104:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
tss:x:105:105:TPM software stack,,,:/var/lib/tpm:/bin/false
systemd-oom:x:990:990:systemd Userspace OOM Killer:/:/usr/sbin/nologin
kernoops:x:106:65534:Kernel Oops Tracking Daemon,,,:/usr/sbin/nologin
whoopsie:x:107:109::/nonexistent:/bin/false
```


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```
root@abdullah-sohail-VMware-Virtual-Platform:/home/abdullah-sohail# passwd -S alice
alice P 2026-03-01 0 99999 7 -1
root@abdullah-sohail-VMware-Virtual-Platform:/home/abdullah-sohail# grep alice /etc/passwd
alice:x:1500:1002::/home/alice:/usr/sbin/nologin
root@abdullah-sohail-VMware-Virtual-Platform:/home/abdullah-sohail#
```

```
alice:x:1500:1002::/home/alice:/usr/sbin/nologin
root@abdullah-sohail-VMware-Virtual-Platform:/home/abdullah-sohail# chage -l alice
Last password change           : Mar 01, 2026
Password expires                : never
Password inactive              : never
Account expires                : never
Minimum number of days between password change : 0
Maximum number of days between password change : 99999
Number of days of warning before password expires : 7
root@abdullah-sohail-VMware-Virtual-Platform:/home/abdullah-sohail#
```

```
root@abdullah-sohail-VMware-Virtual-Platform:/home/abdullah-sohail# ls -ld /home/alice
drwxr-x--- 14 alice alice 4096 Mar  1 12:41 /home/alice
root@abdullah-sohail-VMware-Virtual-Platform:/home/abdullah-sohail# usermod -g newgroup ah
```

16. Configuration Best Practices

- Discuss the security implications of enabling direct root login on a Linux system.
- What are the best practices for managing user and group permissions in a multi-user Linux environment?

Direct root login risks

Problems:

- Very dangerous if password leaks
- No user accountability
- Brute-force attacks target root
- Hard to audit actions

Better approach:

- Disable root login
 - Use **sudo** instead.
-

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Best practices for user management

1. Use **least privilege principle**
2. Avoid direct root login
3. Use **groups instead of individual permissions**
4. Remove unused users
5. Use **strong passwords**
6. Audit permissions regularly
7. Use **ACLs when needed**